
Final Year Project

LiDAR Re-localisation for Resource-Constrained Devices: A Simulation-Based Analysis

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Abstract

While LiDAR sensors are key to navigation in autonomous vehicles, their use in smaller devices like mobile phones or IoT gadgets faces hurdles due to cost and computational limits. This project explores whether reliable LiDAR-based re-localisation – figuring out where a device is within a known map – is practical for these resource-constrained platforms. Instead of using actual low-cost sensors, I simulated their limitations by taking high-quality data from the well-known KITTI automotive dataset and systematically degrading it.

I applied several types of degradation to the KITTI point clouds to mimic the characteristics of cheaper sensors: reducing the maximum sensing range, limiting the field of view, decreasing point density through voxel downsampling, and adding artificial noise. Using this simulated low-quality data, I tested the performance of a standard re-localisation approach. Specifically, I built a reference map from the original KITTI data and then attempted to locate a subsequent degraded query scan within that map using the Iterative Closest Point (ICP) algorithm.

By comparing the position and orientation estimated by ICP against the ground truth from the dataset, I quantified the impact of each degradation type. My experiments systematically varied the severity of range limits, field of view restrictions, downsampling levels, and noise intensity, measuring the resulting translation and rotation errors. The findings highlight which sensor limitations most significantly challenge ICP-based re-localisation and provide insights into the potential accuracy trade-offs when deploying such techniques on mobile or IoT devices with less capable LiDAR sensors. This simulation-based analysis helps assess the feasibility of extending LiDAR localisation beyond high-end automotive applications.

Repository of the project (GitLab link): [Gitlab](#)

Chapter 1: Project Specification

This chapter reiterates the primary objectives established at the outset of this project. These goals, divided into Core and Advanced categories, serve as the benchmark for evaluating the research conducted using the KITTI dataset for outdoor environment simulation. They define the scope of the investigation into LiDAR re-localisation feasibility for resource-constrained devices based on this specific dataset context.

1.1 Core Goals

Survey of LiDAR Re-localization Techniques

- Conduct a thorough literature review of state-of-the-art LiDAR SLAM and re-localization algorithms.
- Identify common computational demands, sensor assumptions, and domain constraints.
- Establish a clear baseline of methods that can be readily adapted to lower-power devices.

Preliminary Feasibility Assessment

- Investigate the specific limitations of embedded platforms, including CPU/GPU capacity, memory footprint, and power consumption.
- Determine algorithmic components (e.g., feature extraction, loop-closure detection) most sensitive to hardware constraints.
- Highlight potential adaptations or optimizations for real-time LiDAR processing in resource-limited settings.

Proof-of-Concept Implementation

- Develop a lightweight LiDAR re-localisation pipeline, integrating open-source or in-house modules.
- Evaluate initial performance (accuracy, latency, and scalability) on representative embedded hardware or low-power computing devices.
- Document any bottlenecks or failure cases for further investigation.

1.2 Advanced Goals

Algorithmic Optimization and Sensor Fusion

- Tailor existing re-localization methods to operate on noisier or sparser data from budget-friendly LiDAR sensors.
- Investigate fusion with additional onboard sensors (IMU, odometry) to improve robustness and accuracy.
- Propose novel approaches (e.g., reduced-resolution mapping, selective point-cloud sampling) that maintain localization performance under strict resource constraints.

Comprehensive Validation

- Conduct extensive benchmarking across diverse scenarios (indoor, outdoor, dynamic environments) using multiple low-cost LiDAR models.
- Compare results with high-end automotive LiDAR setups to quantify trade-offs in accuracy versus computational load.
- Generate metrics such as frame rate, CPU/GPU usage, and drift over distance to guide future design decisions.

Deployment and Wider Applicability

- Demonstrate real-time operation on at least one commercial grade mobile/IoT device, illustrating practical integration challenges.
- Explore the scalability of the system for broader consumer applications, including home robotics, AR/VR, and personal navigation.
- Identify key open problems where further research or hardware advancements might be necessary for widespread adoption.

Chapter 2: Introduction

This project delves into the world of spatial awareness for technology, specifically looking at how devices can figure out where they are using LiDAR sensors. While we often see impressive navigation in self-driving cars using high-end LiDAR, the question arises: can we bring similar capabilities to the smaller, everyday devices we use, like mobile phones or simple robots used in homes or warehouses? These smaller devices operate under much tighter constraints – they don't have the powerful processors or expensive sensors found in autonomous vehicles. My work explores the feasibility of using LiDAR for re-localisation (determining position within a known map) on these resource-constrained platforms. The core idea is to understand how well existing techniques hold up when faced with the kind of limited sensor data and processing power typical of mobile and IoT devices.

The motivation behind this investigation is twofold. Firstly, while cameras are common for localisation (think smartphone AR), there are many situations where cameras aren't ideal due to privacy concerns (like in homes or hospitals) or poor lighting conditions. LiDAR offers a way around this by directly measuring geometry, regardless of light. Secondly, enabling reliable localisation on affordable, low-power devices could unlock a wave of new applications. Imagine smarter home robots that navigate efficiently, better indoor navigation systems on our phones, or even simple IoT devices that understand their spatial context. However, bridging the gap between powerful, expensive automotive LiDAR systems and these constrained devices is a significant challenge. This project aims to assess this gap by simulating the limitations of low-cost sensors using high-quality data from automotive benchmarks like KITTI. By understanding how performance degrades under these simulated conditions, we can better gauge the practical potential and challenges of bringing robust LiDAR localisation to the devices around us.

Chapter 3: Related Work and Ideas

To effectively situate this project, a comprehensive understanding of the existing landscape in Simultaneous Localisation and Mapping (SLAM), particularly concerning LiDAR-based approaches and their application constraints, is necessary. SLAM addresses the fundamental challenge faced by an autonomous agent: concurrently building a map of an unknown environment while determining its own pose (position and orientation) within that map [1]. This capability underpins autonomous navigation, enabling operation in environments where external positioning systems like GPS may be unavailable or unreliable. Figure 3.1 provides a high-level illustration of this process.

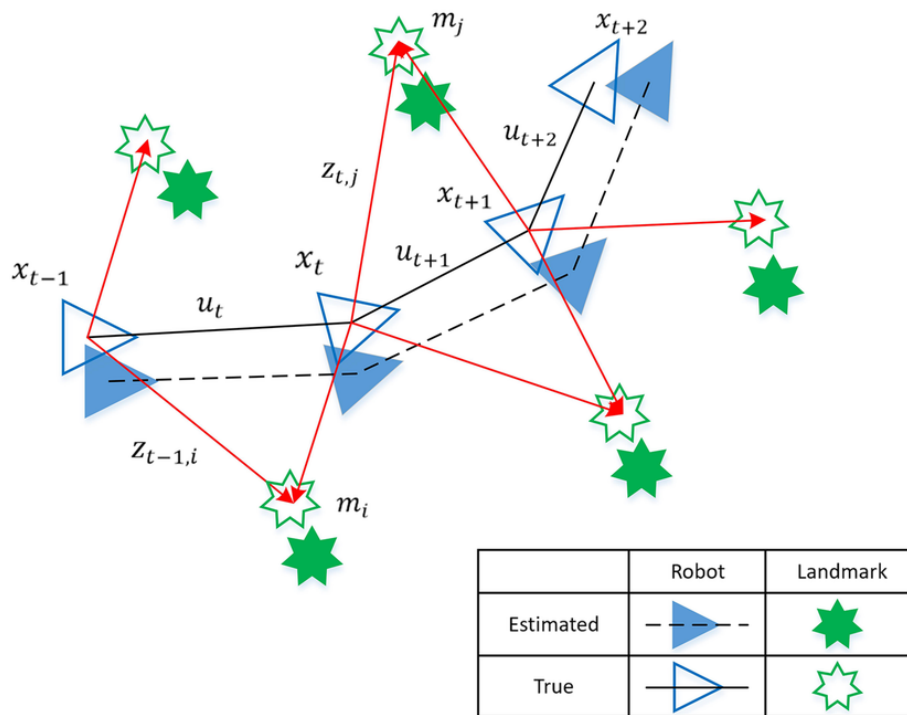


Figure 3.1: Conceptual overview of Simultaneous Localisation and Mapping (SLAM), highlighting the interplay between map creation and pose estimation.

3.1 Evolution from Visual SLAM to LiDAR SLAM

Historically, much pioneering work in SLAM utilized cameras as the primary sensor modality, giving rise to Visual SLAM (V-SLAM) [2]. The affordability and information richness of cameras made them an attractive option. V-SLAM encompasses various techniques, broadly categorized as:

- **Feature-Based Methods:** These detect sparse, salient points (keypoints) in images (e.g., corners using SIFT, SURF, ORB) and match these keypoints across consecutive frames to estimate camera motion and triangulate 3D point positions. ORB-SLAM is a prominent example in this category.

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- **Direct Methods:** These methods operate directly on pixel intensities, minimizing photometric errors between frames without explicit feature detection. Examples include LSD-SLAM (Large-Scale Direct Monocular SLAM). They can potentially utilize more image information but may be more sensitive to illumination changes.
 - **Semi-Direct Methods:** These combine aspects of both, often tracking features detected initially but using direct methods for refinement (e.g., SVO - Semi-direct Visual Odometry).

While V-SLAM has achieved significant success, particularly in augmented reality and robotics applications within well-structured environments [2], it suffers from inherent limitations. Performance degrades significantly under challenging illumination conditions (e.g., low light, high dynamic range, glare). Textureless surfaces (like white walls) provide few features for matching, and rapid motion can cause image blur, hindering tracking. Crucially, the reliance on cameras raises significant privacy concerns, limiting deployment in sensitive environments such as homes, hospitals, or secure industrial sites [3].

These challenges motivated the exploration and adoption of LiDAR (Light Detection and Ranging) sensors. LiDAR systems actively emit laser pulses and measure the time-of-flight (or phase shift) of the reflected signals to determine precise distances to surrounding objects. This process generates a direct 3D representation of the environment as a point cloud, typically composed of thousands or hundreds of thousands of points, each with (x, y, z) coordinates and sometimes reflection intensity. Key advantages of LiDAR include:

- **Direct Depth Measurement:** Provides accurate geometric information directly, unlike monocular V-SLAM which struggles with scale ambiguity.
- **Lighting Invariance:** As an active sensor, LiDAR performance is largely unaffected by ambient lighting conditions, enabling reliable operation day or night.
- **Geometric Richness:** The dense point clouds capture detailed 3D structure crucial for navigation and mapping.

Consequently, LiDAR has become a cornerstone sensor for autonomous vehicles [4] and increasingly for various robotics applications [5].

3.2 Core Techniques in LiDAR-Based SLAM

LiDAR SLAM algorithms primarily rely on registering incoming point clouds against previous scans or a maintained map to estimate the sensor's motion and update the map. Within the broader SLAM ecosystem, it's important to distinguish key components:

- **LiDAR Odometry (LO):** Focuses on estimating the *relative* motion between consecutive time steps, often by matching the current scan to the previous one (scan-to-scan) or a local map of recent scans (scan-to-map) [6]. LO provides high-frequency updates but accumulates drift over time.
- **Mapping:** Involves integrating multiple aligned scans, often using their estimated poses from odometry, to build a consistent representation of the environment (e.g., a point cloud map, voxel grid, or feature map) [1].

- **Place Recognition / Loop Closure:** This involves detecting when the sensor revisits a previously mapped area. A successful detection (loop closure) allows the system to add constraints to an optimization framework (like pose graph optimization) to correct accumulated drift and improve global map consistency [1].
- **Re-localisation:** This is the specific problem of determining the device's *absolute* 6-DOF pose (position and orientation) within a *pre-existing* map, given a current LiDAR scan [6]. It can be used for initial localisation ("Where am I?") or recovery after tracking failure (the "kidnapped robot problem"), and it fundamentally relies on effective scan-to-map matching.

This project focuses specifically on the **re-localisation** problem, assessing the feasibility of the underlying scan-to-map matching step under simulated mobile/IoT constraints.

3.2.1 Scan-to-Map Matching Algorithms

The core of re-localisation is aligning the current scan (source) to the map (target). Key algorithms include:

- **Iterative Closest Point (ICP):** This classic algorithm [6] provides a versatile framework for aligning two point sets. The basic process involves: (1) For each point in the source cloud, find its closest corresponding point in the target cloud. (2) Compute the rigid transformation (rotation $\mathbf{R}$ and translation $\mathbf{t}$) that minimizes an error metric based on the found correspondences. A common metric is the sum of squared distances: $\min_{\mathbf{R}, \mathbf{t}} \sum_i \|(\mathbf{R}p_i + \mathbf{t}) - q_i\|^2$. This minimization has a closed-form solution using Singular Value Decomposition (SVD).
- **Apply Transformation:** Update the source points: $\mathcal{P} \leftarrow \{\mathbf{R}p_i + \mathbf{t}\}$.
- **Iterate:** Repeat steps 1-3 until convergence (e.g., change in error metric or transformation is below a threshold, or a maximum number of iterations is reached).

ICP Variants and Considerations:

- **Point-to-Plane ICP:** Instead of minimizing point-to-point distance, this variant minimizes the distance between each source point and the plane containing its corresponding target point (estimated using the target point's normal vector) [7, 8]. This often leads to faster convergence and higher accuracy in environments with planar structures. The error metric becomes $\min_{\mathbf{R}, \mathbf{t}} \sum_i \|((\mathbf{R}p_i + \mathbf{t}) - q_i) \cdot \mathbf{n}_i\|^2$, where $\mathbf{n}_i$ is the normal at q_i . This minimization typically requires iterative numerical methods rather than a closed-form solution.
- **Robustness:** Standard ICP is sensitive to outliers and requires good overlap. Trimmed ICP (TrICP) discards a percentage of correspondences with the largest errors. Generalized-ICP (G-ICP) models uncertainty and minimizes a plane-to-plane metric [6]. Robust cost functions can also be used during minimization.
- **Initialization:** ICP is a local optimization method and requires a reasonably accurate initial guess for the transformation to converge to the correct global minimum [6]. Poor initialization can lead to convergence to an incorrect alignment.
- **Computational Cost:** The nearest neighbour search (Step 1) is typically the bottleneck, with complexity roughly $O(N_p \log N_q)$ using efficient spatial data structures like k-d trees, where N_p and N_q are the number of points in the source and target clouds, respectively.

This project utilizes the point-to-plane ICP implementation within Open3D [9] for its balance of accuracy and availability.

Normal Distributions Transform (NDT)

NDT offers a different philosophy, avoiding explicit point correspondences [10]. It operates as follows:

1. **Voxelization:** The space occupied by the target point cloud $\mathcal{Q}$ is divided into a regular grid of voxels (usually cubes).
2. **Distribution Fitting:** For each voxel containing a sufficient number of points (e.g., > 5), the mean μ and covariance matrix Σ of the points within it are calculated. This defines a 3D Gaussian distribution $N(\mu, \Sigma)$ representing the local surface structure within that voxel.
3. **Optimization:** Given a source point cloud $\mathcal{P}$ and a current transformation estimate $(\mathbf{R}, \mathbf{t})$, the algorithm evaluates how well the transformed source points $p'_i = \mathbf{R}p_i + \mathbf{t}$ fit the Gaussian model of the target map. This is often formulated as maximizing a likelihood score, which involves summing the probability density function (PDF) values of the Gaussians at the locations of the transformed source points: $\text{maximize}_{\mathbf{R}, \mathbf{t}} \sum_i \exp(-\frac{1}{2}(p'_i - \mu_{k(i)})^T \Sigma_{k(i)}^{-1} (p'_i - \mu_{k(i)}))$, where $k(i)$ indicates the voxel containing p'_i . This maximization is typically performed using iterative optimization methods like Newton-Gauss or Levenberg-Marquardt.

NDT Variants and Considerations:

- **Efficiency:** NDT avoids the $O(N_p \log N_q)$ nearest neighbour search of ICP. The cost is dominated by building the voxel grid model and the iterative optimization step.
- **Robustness:** By representing surfaces statistically, NDT can be less sensitive to noise and variations in point density than basic ICP [11].
- **Parameters:** Performance is highly dependent on the chosen voxel size. Too large, and details are lost; too small, and computation increases while robustness to sparsity might decrease [12].
- **Accuracy:** While robust, NDT might not achieve the same level of fine-grained accuracy as point-to-plane ICP in highly structured environments [13]. Variants like multi-resolution NDT exist.

3.2.2 Feature-Based SLAM (LOAM Family)

Methods like LOAM [14] and LeGO-LOAM [5] represent a different paradigm focused on computational efficiency. They operate in stages:

1. **Feature Extraction:** Analyze local point neighbourhoods along scan lines (or in local regions) to identify points with high curvature (edge features) and low curvature (planar features). Ground points are often segmented and treated separately or removed.
2. **Odometry Estimation:** Estimate motion between consecutive scans by matching features (edge-to-edge line distances, point-to-plane distances for planar features). This matching involves fewer points than full ICP/NDT.
3. **Mapping:** Integrate the features from multiple scans into a global feature map using the estimated poses.
4. **Loop Closure (Optional):** Detect loop closures, often using ICP or feature matching between the current scan and the global feature map, followed by pose graph optimization.

Figure 3.2 conceptually shows feature extraction. The main advantage is the significant reduction in the number of points processed for registration and optimization. However, the reliance on robust feature extraction is also a potential weakness, especially with degraded sensor data [15].

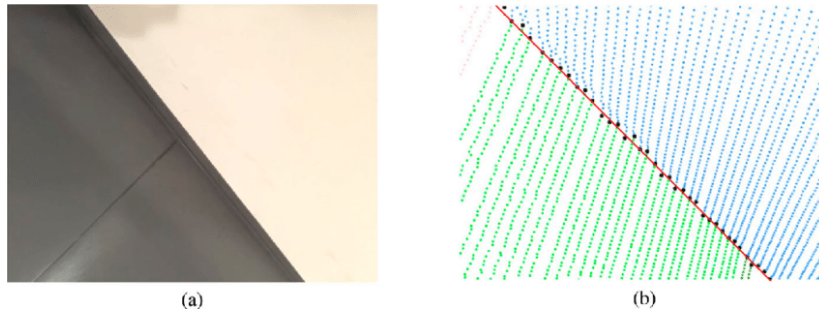


Figure 3.2: Conceptual illustration of edge and plane feature extraction in LOAM-style methods.

3.2.3 Place Recognition and Re-Localisation

Robustly determining location within a large, previously built map (re-localisation) often requires more than just local scan matching. Place recognition techniques aim to find candidate locations in the map that match the current view.

- **Descriptor-Based:** Scan Context++ [16] creates a 2D descriptor encoding the distribution of points around the sensor, designed to be robust to viewpoint rotation. Matching current descriptors against a database of map descriptors allows efficient retrieval of potential location matches.
- **Deep Learning:** OverlapNet [7] trains a neural network to predict the geometric overlap between LiDAR scans, providing a similarity score useful for identifying loop closures or re-localisation candidates. While powerful, computational cost is a major consideration for mobile deployment.

Once a candidate pose is retrieved via place recognition, fine alignment is typically performed using ICP or NDT between the current scan and the relevant submap. My work simulates this fine alignment step, assuming a coarse initial pose is available (provided by the ground truth in the simulation).

3.3 The Challenge: Resource Constraints

The transition from high-resource AV platforms to resource-constrained mobile/IoT devices introduces significant challenges. The algorithms and assumptions prevalent in AV research, often validated using benchmarks like KITTI [4], nuScenes [17], Waymo [18], Argoverse [19], and Ford [20], do not directly translate [21].

3.3.1 Sensor Limitations (The "Sensor Gap")

As detailed in Table 3.1, sensors viable for mobile/IoT differ drastically from high-end AV LiDARs across multiple dimensions: cost, power consumption, size/weight, maximum range, field of view

(FoV), point density/resolution, and scan pattern [21–23]. Mobile sensors like the Livox Mid-40 [24] or RoboSense E1R [25] typically offer shorter ranges, limited FoVs, and sparser point clouds compared to a Velodyne HDL-64E used in the original KITTI dataset [26]. This "sensor gap" means algorithms must cope with fundamentally less informative input data.

Table 3.1: Comparison of AV Benchmark LiDAR vs. Potential Mobile/IoT LiDAR Specifications (Adapted from background research [14, 21–23])

Parameter	AV Benchmark LiDARs (Examples: Velodyne HDL-64E/32E, Ouster OS1)	Potential Mobile/IoT LiDARs (Examples: Livox Mid-40/Avia, RoboSense E1R, RPLIDAR, MEMS-based)
Type	Mechanical Spinning	Mechanical Spinning (low beam), Solid-State (MEMS, Flash, OPA), Hybrid
Cost Tier	Very High (\$k+)	Low to Medium (\$ hundreds - \$ few k)
Power Tier	High (Tens of W)	Low to Medium (<5W - 30W)
Typical Range	Long (100m - 200m+)	Short to Medium (e.g., 10m - 75m)
Typical Horiz. FoV	Wide (360°)	Often Limited (e.g., 60°–120°, some 360°)
Typical Vert. FoV	Moderate (20°–40°)	Variable, often limited (20°–60°, some wider)
Typical Angular Resolution	High (0.1°–0.4°)	Variable, potentially lower
Scan Pattern	Regular Scan Lines	Regular Lines (spinning), Non-repetitive (e.g., Livox), Raster (MEMS)
Size/Weight Tier	Large / Heavy	Small / Light

3.3.2 Computational Limits

Mobile/IoT devices typically feature low-power ARM-based CPUs with limited cores and clock speeds, minimal RAM (often just a few GBs or less), and potentially no dedicated GPU or only a low-power integrated one [22]. This severely restricts the complexity of algorithms that can run in real-time. Energy efficiency is also paramount for battery-powered operation. Algorithms requiring large map storage or complex optimization routines (like bundle adjustment or large-scale pose graph optimization) may be entirely infeasible.

3.4 Existing Research Gaps and Project Focus

While research exists on mobile SLAM, often focusing on vision or 2D LiDAR [22], a systematic evaluation of adapting established 3D LiDAR re-localisation techniques to mobile constraints using standard benchmarks is less common. This project aims to fill this gap via simulation.

My methodology involves:

1. **Using the KITTI Dataset:** Leveraging the complexity and ground truth of the KITTI dataset [4] as a foundation. Figure 3.3 shows an example point cloud.

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2. **Simulating Degradation:** Applying controlled degradation (range/FoV limits, downsampling, noise) to KITTI data to mimic low-cost sensor output.
 3. **Evaluating ICP Performance:** Assessing point-to-plane ICP's [9] re-localisation accuracy (translation/rotation error) under these simulated constraints against a map built from original data.

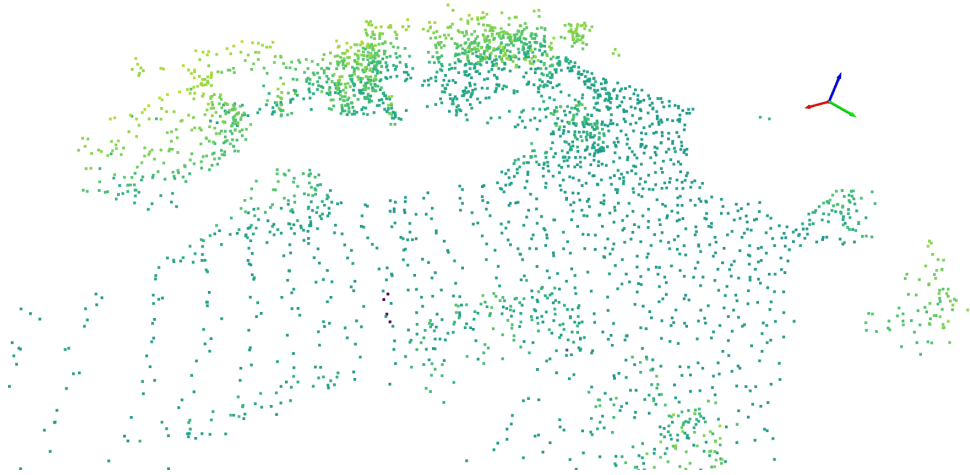


Figure 3.3: Example Velodyne point cloud from the KITTI dataset, illustrating typical density and structure.

This simulation-based feasibility study allows for reproducible analysis of how core techniques perform when faced with data characteristic of resource-constrained devices, providing insights into the challenges and potential of mobile LiDAR re-localisation without requiring immediate hardware prototyping. It focuses on quantifying the performance impact of specific sensor limitations on a baseline algorithm (ICP).

Chapter 4: Data Considerations

This chapter details the data utilized in this project, addressing its nature, source, processing, and relevant practical considerations. The core of this research relies on manipulating an existing, high-quality dataset to simulate the conditions relevant to the project's goals.

4.1 Nature and Scale of the Data

The primary data source for this project is a single sequence from the KITTI raw dataset [4], specifically the drive recorded on 2011_09_26, designated as drive '0035'. This dataset represents a real-world outdoor driving scenario captured using high-end sensors.

The key data components utilized are:

- **LiDAR Point Clouds:** Generated by a Velodyne HDL-64E sensor. For the specific data extract I used, these were provided as text files ('.txt') for each frame (scan), where each line contains the x, y, z coordinates and intensity value for a single point. Each frame contains approximately 117,000 to 119,000 points. The sequence '0035' contains 1408 frames in total, captured at 10 Hz.
- **Ground Truth Poses:** Derived from the OXTS (GPS/IMU) data provided with the KITTI raw dataset. This data includes latitude, longitude, altitude, roll, pitch, and yaw measurements for each frame, which I processed to obtain 4x4 ground truth pose matrices representing the vehicle's trajectory in a local Cartesian coordinate frame.
- **Calibration Data:** Files providing the extrinsic transformations between sensors (e.g., 'calib_imu_to_velo.txt', 'calib_velo_to_cam.txt') and intrinsic camera parameters (though cameras were not directly used in the final simulation).

The data is entirely **static** and was downloaded and stored **locally** on the development machine for offline processing. No real-time streaming or remote data access was involved. The scale, while significant for a single sequence (over 1400 frames), was manageable for local processing and simulation.

4.2 Source and Rationale

The KITTI dataset [4] was chosen as the data source for several key reasons:

- **Public Availability and Standardization:** It is a widely recognized and publicly accessible benchmark in the autonomous driving and robotics community, facilitating reproducible research.

- **High-Quality Ground Truth:** It provides accurate ground truth poses derived from a high-grade GPS/IMU system, which is essential for evaluating the accuracy of the localisation algorithms.
- **Rich Sensor Data:** The original data comes from a high-resolution LiDAR sensor, providing a dense, high-quality baseline point cloud that is suitable for applying various simulated degradations.
- **Suitability for Simulation:** Using a high-quality dataset allows for controlled degradation to simulate different levels of sensor limitations (range, FoV, density, noise). This approach avoids the need for acquiring and calibrating multiple physical low-cost sensors and collecting new data, while still allowing for quantitative analysis based on real-world geometry.

While KITTI is an automotive dataset capturing outdoor driving scenes, and the project goal relates to mobile/IoT devices which often operate indoors, using KITTI was deemed appropriate for this initial feasibility study. The primary aim was to understand how standard algorithms (ICP) behave when input data quality is reduced in specific ways, using a dataset with reliable ground truth. The complex geometric structures and dynamic elements present in KITTI provide a challenging baseline for evaluating the degradation effects, even if the specific environmental context differs from some target mobile/IoT applications. No Irish datasets of comparable quality and public availability for this specific task were identified.

4.3 Data Preprocessing and Simulation

The raw KITTI data required several preprocessing and simulation steps to be used in the experiments:

1. **Data Loading:** Custom Python scripts were developed to parse the text-based LiDAR scan files, the OXTS data packets, and the calibration files for the specific sequence used.
2. **Pose Calculation:** The raw OXTS latitude, longitude, altitude, and orientation data was converted into a sequence of 4x4 homogeneous transformation matrices representing the ground truth pose trajectory relative to the starting position of the sequence, using a Mercator projection for coordinate conversion.
3. **Map Generation (for Re-localisation):** A reference map was created by taking the original, non-degraded point clouds from the first few frames (frames 0-4), transforming them into the world coordinate frame using their respective ground truth poses, and concatenating them into a single large point cloud. This combined map was then downsampled using a voxel grid (voxel size 0.2m) to create the final target map for re-localisation, containing approximately 83,000 points.
4. **Query Frame Degradation (Simulation):** The point cloud for the chosen query frame (frame 15) was subjected to a sequence of simulated degradations to mimic a lower-cost sensor:
 - *Range Limiting:* Points beyond a maximum distance (e.g., 40m) from the sensor origin were removed.
 - *Field of View (FoV) Limiting:* Points outside a specified horizontal angular range (e.g., -60° to $+60^\circ$ relative to the forward X-axis) were removed.

-
- *Voxel Downsampling*: The remaining points were downsampled using a voxel grid (e.g., voxel size 0.3m, 0.5m, 1.0m etc.) to reduce density.
 - *Noise Injection*: Gaussian noise (e.g., standard deviation 0.1m) was added to the x, y, z coordinates of the points.

The parameters for these degradations were varied systematically across different experimental runs.

This process resulted in a high-quality reference map and various degraded query point clouds suitable for testing the re-localisation algorithm under simulated constraints.

4.4 Licensing and Availability

The KITTI dataset is publicly available for non-commercial research purposes under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 License. My use of the data for this academic project complies with these terms. The original KITTI data can be accessed from the official website (<http://www.cvlibs.net/datasets/kitti/>).

The degraded point clouds and specific map generated within this project are derived directly from the KITTI data. While the scripts used for processing could potentially be shared, the derived data itself is not planned for public release as it can be regenerated from the original public dataset using the described methodology.

4.5 Privacy and Ethical Considerations

This project utilizes existing, publicly released data collected in public road environments. While the original dataset collection involved driving in public spaces, my research focused solely on processing the anonymized geometric LiDAR point cloud data and the associated GPS/IMU readings. No camera images were used, and the point cloud data itself does not contain directly identifiable personal information. The simulation methodology further abstracts from the raw data. Therefore, the data usage in this project poses minimal privacy or ethical concerns. Any future work involving deployment on real devices, especially in private spaces, would require a thorough re-evaluation of privacy implications and ethical guidelines.

Chapter 5: Outline of Approach

This chapter details the specific methodology employed in this project to investigate the feasibility of LiDAR re-localisation on resource-constrained devices through simulation. The approach focuses on using existing high-quality data and degrading it systematically to emulate lower-cost sensor performance, followed by evaluating a standard localisation algorithm.

5.1 Overall Methodology

The core approach involved the following steps, executed using custom Python scripts leveraging standard libraries:

1. Dataset Selection and Preparation:

- The KITTI raw dataset [4] was selected due to its public availability, real-world complexity, and accurate ground truth.
- Specifically, the sequence '2011_09_26', drive '0035' (an outdoor residential/urban scenario) was used for all experiments.
- Data components (LiDAR scans as '.txt' files, OXTS packets, calibration files) were loaded using custom parsing functions.

2. Ground Truth Pose Calculation:

- The raw OXTS sensor readings (latitude, longitude, altitude, roll, pitch, yaw) for the entire sequence were processed.
- A function implementing Mercator projection and Euler angle rotations (ZYX convention) was used to convert these readings into a sequence of 4x4 homogeneous pose matrices ($T_{world \leftarrow sensor}$) relative to the starting frame's pose, providing the ground truth trajectory.

3. Map Generation:

- A reference map for the re-localisation task was built using the *original*, non-degraded LiDAR point clouds from the initial frames of the sequence (frames 0 through 4).
- Each frame's point cloud was transformed into the world coordinate frame using its corresponding ground truth pose.
- These transformed point clouds were concatenated into a single map point cloud.
- The resulting map was downsampled using a voxel grid filter (voxel size 0.2m) provided by Open3D [9] to create the final, moderately dense target map for ICP.

4. Simulation of Sensor Degradation:

- A specific query frame (frame 15) was selected for the re-localisation experiments.
- The original point cloud for this frame was subjected to a sequence of simulated degradations to mimic potential low-cost sensor limitations:

- (a) *Range Limiting*: Points beyond a maximum spherical distance from the sensor origin were removed. Tested ranges included 60m, 50m, 40m (default), 30m, 15m, 5m.
- (b) *Field of View (FoV) Limiting*: Points outside a specified horizontal angular range relative to the sensor's forward X-axis were removed. Tested total FoVs included 180° (±90), 120° (±60, default), 90° (±45), 60° (±30).
- (c) *Voxel Downsampling*: The remaining points were downsampled using an Open3D voxel grid filter to reduce density. Tested voxel sizes included 0.1m, 0.3m (default), 0.5m, 0.7m, 1.0m, 1.3m, 1.5m.
- (d) *Noise Injection*: Gaussian noise was added to the x, y, z coordinates of the points using 'numpy.random.normal'. Tested standard deviations included 0.0m, 0.05m, 0.1m (default), 0.2m, 0.3m. A fixed random seed ('np.random.seed(0)') was used to ensure reproducibility of noise across runs.

- In the systematic experiments, only one degradation parameter was varied at a time, while others were held at their default values (Range=40m, FoV=120°, Voxel=0.3m, Noise=0.1m).

5. Re-localisation Algorithm Implementation:

- The Iterative Closest Point (ICP) algorithm was chosen as the core method for scan-to-map matching to estimate the query frame's pose.
- The point-to-plane variant of ICP, as implemented in the Open3D library ('o3d.pipelines.registration.registration_icp' with 'TransformationEstimationPointToPlane'), was used due to its common use and potential for higher accuracy in structured environments compared to point-to-point ICP.
- The degraded query frame point cloud served as the 'source', and the generated map point cloud served as the 'target'.
- The ground truth pose of the query frame was used as the initial guess ('initial_transform') for the ICP algorithm to simulate a scenario where a coarse prior estimate might be available (e.g., from GPS or odometry).

6. Evaluation:

- The 4x4 pose matrix returned by the ICP algorithm ($T_{map \leftarrow query_est}$) was taken as the estimated pose of the query frame relative to the map's world frame.
- This estimated pose was compared against the ground truth pose of the query frame ($T_{map \leftarrow query_gt}$), also relative to the world frame.
- The error was quantified by calculating the relative error transformation: $T_{err} = T_{gt \leftarrow map} \times T_{map \leftarrow est} = (T_{map \leftarrow gt})^{-1} \times T_{map \leftarrow est}$.
- *Translation Error* was calculated as the Euclidean norm of the translation vector in T_{err} 's last column.
- *Rotation Error* was calculated as the angle component of the axis-angle representation derived from the 3x3 rotation matrix part of T_{err} , using functions from 'scipy.spatial.transform'.
- These errors were recorded for each experimental run where a different degradation parameter was varied.
- Visualization of the map, the degraded cloud, and the final alignment was performed using Open3D. Trajectory plots for the odometry simulation were generated using Matplotlib [27].

5.2 Algorithm and Parameter Choices

- **Core Algorithm:** Point-to-Plane ICP (via Open3D [9]). This was chosen as a well-understood baseline algorithm commonly used for precise registration.
- **ICP Parameters:**
 - *Threshold (Max Correspondence Distance):* Set to 0.5 meters ('icp_threshold = 0.5'). This value balances capturing relevant correspondences while rejecting some outliers.
 - *Max Iterations:* Set to 100 ('icp_max_iter = 100') for the re-localisation task to allow sufficient convergence opportunity when aligning to the larger map. (Note: 50 iterations were used in the earlier odometry simulation).
 - *Initial Guess:* The ground truth pose of the query frame was used as the initial transformation estimate provided to ICP. This represents a best-case scenario for ICP initialization.
- **Degradation Defaults (when varying others):**
 - 'max_range_sim = 40.0' meters
 - 'min_fov_sim, max_fov_sim = -60.0, 60.0' degrees (120° total)
 - 'voxel_size_sim = 0.3' meters
 - 'noise_std_dev_sim = 0.1' meters
- **Map Generation:** Frames 0-4 used, map downsampled with 'map_voxel_size = 0.2' meters.
- **Query Frame:** Frame 15 used for re-localisation tests.

5.3 Software Toolchain

The implementation and experiments were conducted using:

- **Programming Language:** Python 3.
- **Core Libraries:**
 - NumPy [28] for numerical computation and array manipulation.
 - Open3D [9] for point cloud data structures, voxel downsampling, ICP implementation, and 3D visualization.
 - SciPy [29] (specifically 'scipy.spatial.transform.Rotation') for rotation matrix calculations and error computation.
 - Matplotlib [27] for generating 2D plots of results and trajectories.

This approach allowed for a systematic simulation and evaluation of a standard re-localisation technique under various controlled degradation conditions, providing quantitative data on the impact of simulated low-cost sensor limitations.

Chapter 6: Project Workplan

This chapter outlines the plan followed for executing this project from its inception around the beginning of October 2024 through to the final report submission deadline of May 1st, 2025. It details the major phases, key tasks, dependencies, and the evaluation strategy employed to assess the project's outcomes.

6.1 Work Plan Overview

The project was structured into several distinct phases, ensuring a logical progression from background research and setup to experimentation, analysis, and final reporting. While some tasks overlapped, the general flow allowed for iterative refinement based on earlier findings. The approximate timeline for each phase was as follows:

- **Phase 1: Literature Review & Foundational Setup (Approx. October - Mid November)**

- Task 1.1: Comprehensive review of literature on LiDAR SLAM, re-localisation techniques (ICP, NDT, feature-based), challenges in resource-constrained environments, and relevant datasets (particularly KITTI). (4 weeks)
- Task 1.2: Selection of the specific KITTI raw data sequence ('2011_09_26', drive '0035'). Setup of the Python development environment, including NumPy, Open3D, SciPy, and Matplotlib. Understanding data formats. (2 weeks)
- Task 1.3: Development and testing of initial Python scripts for loading LiDAR scans ('.txt' format), OXTS packets, and calibration files. Basic data visualization using Open3D. (2 weeks)

Dependency: Literature review informed the choice of simulation methodology and baseline algorithm (ICP). Setup was required for all subsequent implementation.

- **Phase 2: Methodology Development & Implementation (Approx. Mid November - Mid January)**

- Task 2.1: Implementation and validation of the function to convert raw OXTS data into 4x4 ground truth pose matrices using Mercator projection and Euler angles. (2 weeks)
- Task 2.2: Implementation of the map generation procedure: loading multiple original frames, transforming points to world frame using ground truth poses, concatenating, and downsampling (voxel size 0.2m). (1 week)
- Task 2.3: Implementation of the sensor degradation functions: range limiting, FoV limiting, voxel downsampling, and reproducible Gaussian noise injection ('np.random.seed(0)'). (3 weeks)
- Task 2.4: Implementation of the re-localisation step using Open3D's point-to-plane ICP. Implementation of error calculation logic (translation and rotation error) comparing ICP pose estimate against ground truth. (2 weeks)

- Task 2.5: Integration of all components into a single, configurable script for running the re-localisation simulation. (1 week)

Dependency: Ground truth poses needed for map generation and final evaluation. Degradation functions needed before running ICP on simulated data. Error calculation needed for evaluation.

- **Phase 3: Experimentation & Data Collection (Approx. Mid January - End February)**

- Task 3.1: Systematically execute the re-localisation simulation script. Vary one degradation parameter (voxel size, max range, FoV, noise std dev) at a time across a defined range, keeping others at default values. (4 weeks)
- Task 3.2: Re-run specific outlier tests (e.g., voxel size 1.0m) to ensure consistency after implementing the random seed. (1 week)
- Task 3.3: Collect and meticulously record the resulting translation and rotation errors for each experimental configuration in tables. (Ongoing during this phase)

Dependency: Requires the completed simulation script from Phase 2.

- **Phase 4: Analysis & Visualization (Approx. March)**

- Task 4.1: Develop Python scripts using Matplotlib to plot the collected error data (Translation/Rotation Error vs. each varied degradation parameter). (2 weeks)
- Task 4.2: Analyze the generated plots and quantitative results. Identify trends, sensitivities, anomalies, and formulate interpretations regarding the impact of each degradation type. (2 weeks)

Dependency: Requires the collected experimental data from Phase 3.

- **Phase 5: Report Writing & Submission (Approx. April - May 1st)**

- Task 5.1: Draft initial chapters: Introduction, Related Work, Data Considerations, Outline of Approach, referencing literature and methodology developed. (2 weeks)
- Task 5.2: Draft core results chapters: Methodology details, Results (including plots and data tables), Discussion (interpreting results based on analysis from Phase 4), and Conclusion. (2 weeks)
- Task 5.3: Finalize bibliography using BibTeX, ensure all figures and tables are correctly formatted and referenced, refine LaTeX formatting. (1 week)
- Task 5.4: Comprehensive review, proofreading for clarity, grammar, and consistency. Final compilation and submission of the PDF report by the deadline. (1 week)

Dependency: Builds upon all previous phases, particularly the analysis and results.

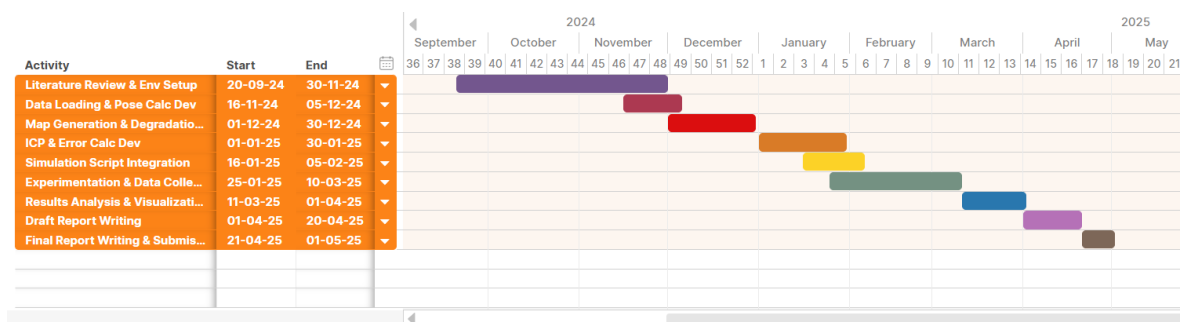


Figure 6.1: Project Timeline Gantt Chart (October 2024 - May 2025).

6.2 Evaluation Plan

The evaluation of this project's success focuses on the quantitative results derived from the simulation experiments, aiming to rigorously assess the impact of simulated sensor limitations on a standard re-localisation algorithm.

- **Core Task:** The central evaluation involves measuring the accuracy of the pose estimated by the point-to-plane ICP algorithm when attempting to re-localise a degraded query frame (frame 15) within a map built from original data (frames 0-4) from the KITTI sequence ('2011_09_26', drive '0035').
- **Primary Metrics:**
 - **Translation Error (meters):** The Euclidean distance between the estimated position and the ground truth position of the query frame. Calculated from the relative error transformation $T_{err} = (T_{map \leftarrow gt})^{-1} \times T_{map \leftarrow est}$.
 - **Rotation Error (degrees):** The angle component of the axis-angle representation of the rotation difference between the estimated orientation and the ground truth orientation. Calculated from the rotation part of T_{err} .
- **Methodology:** The evaluation involved systematically varying individual degradation parameters (maximum range, horizontal FoV, voxel downsampling size, Gaussian noise standard deviation) across a range of values, while holding other parameters constant at defined defaults. For each configuration, the re-localisation simulation was run, and the resulting translation and rotation errors were recorded. The use of a fixed random seed ensured reproducibility of the noise injection and thus the results.
- **Analysis:** The collected error data was plotted using Matplotlib, showing Translation Error vs. Parameter Value and Rotation Error vs. Parameter Value for each of the four degradation types. These plots form the basis for analyzing trends, identifying sensitivities, and understanding how different simulated sensor limitations affect ICP performance.
- **Success Criteria:** The project is considered successful based on:
 1. The successful implementation and execution of the complete simulation pipeline (data loading, pose calculation, map generation, degradation, ICP re-localisation, error calculation).
 2. The generation of quantitative results (translation/rotation errors) for systematic variations in simulated sensor parameters.
 3. A clear analysis and discussion of these results, supported by plots, identifying how different types and levels of degradation impact the baseline re-localisation algorithm's accuracy.

The goal was not necessarily to achieve low error under all conditions, but rather to successfully simulate the process and quantify the performance trade-offs inherent in using degraded sensor data, thereby assessing the feasibility challenges. No qualitative user feedback was planned as the project focused on simulation-based analysis.

Chapter 7: Results

This chapter presents the quantitative results obtained from the re-localisation simulation experiments. The experiments systematically varied key sensor degradation parameters (voxel size, maximum range, field of view, and noise level) applied to a query frame from the KITTI dataset [4]. The accuracy of re-localising this degraded frame against a map built from original data, using the point-to-plane ICP algorithm [9], was measured in terms of translation and rotation error relative to the ground truth pose. The following sections visualize and briefly discuss the impact of each parameter.

7.1 Impact of Voxel Size (Downsampling)

Figure 7.1 illustrates the effect of varying the voxel size used for downsampling the query point cloud, while keeping other degradation parameters constant (Range=40m, FoV=120°, Noise=0.1m std dev).

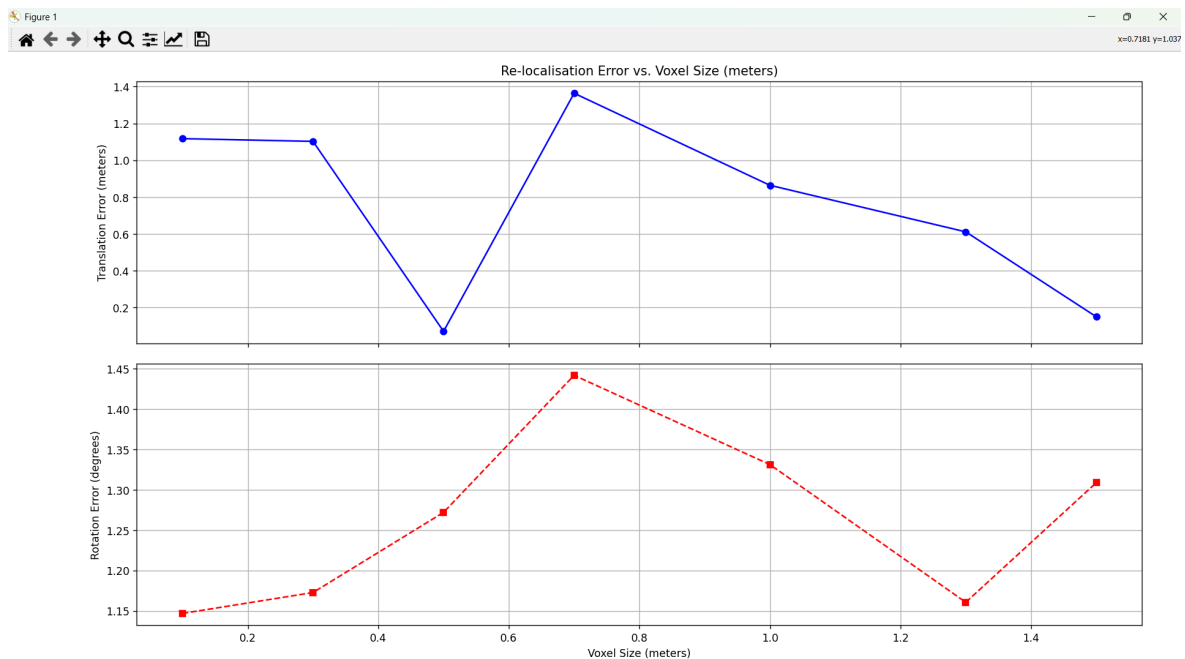


Figure 7.1: Re-localisation Translation and Rotation Error vs. Voxel Size.

As observed in Figure 7.1, the translation error exhibits a non-monotonic trend, decreasing sharply at a voxel size of 0.5m before fluctuating at larger sizes. This suggests a complex relationship where moderate downsampling might remove noise or outliers beneficially for ICP in this specific configuration, while very high downsampling leads to unpredictable results. The rotation error remains relatively stable across the tested voxel sizes, indicating less sensitivity to point density changes for orientation estimation in this case.

7.2 Impact of Maximum Sensor Range

Figure 7.2 shows the re-localisation errors resulting from varying the maximum simulated sensor range, keeping other parameters at their defaults (Voxel=0.3m, FoV=120°, Noise=0.1m std dev).

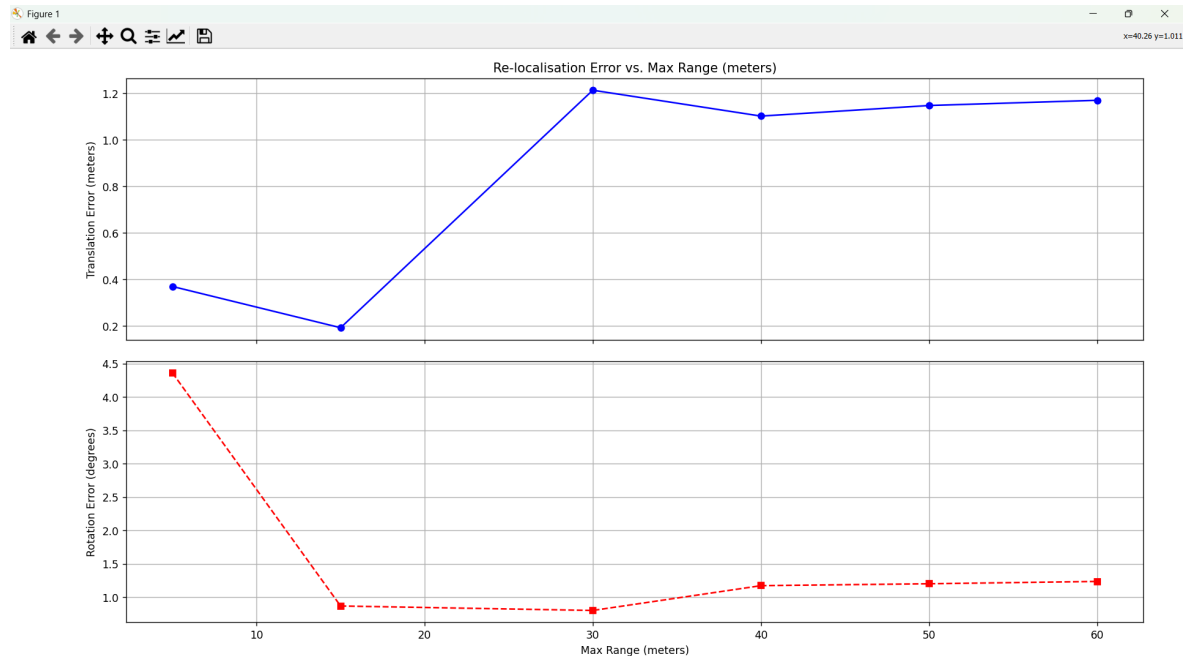


Figure 7.2: Re-localisation Translation and Rotation Error vs. Maximum Sensor Range.

The results in Figure 7.2 indicate that translation error is relatively stable for ranges between 30m and 60m, with an unexpected drop at 15m. The rotation error, however, remains low for ranges down to 15m but increases dramatically at 5m. This highlights the critical importance of sufficient range for robust orientation estimation, as very short ranges severely limit the geometric constraints available to the ICP algorithm.

7.3 Impact of Field of View (FoV)

The effect of limiting the horizontal field of view (FoV) is presented in Figure 7.3. The total horizontal angle was varied, while range (40m), voxel size (0.3m), and noise (0.1m std dev) were kept constant.

Figure 7.3 shows that translation error tends to decrease as the FoV narrows from 180° down to 60°, suggesting that focusing on the central view might benefit positional accuracy in this scenario. Conversely, rotation error is lowest at the widest FoV (180°) and increases significantly at the narrowest FoV (60°), indicating that a wider view provides better constraints for orientation estimation.

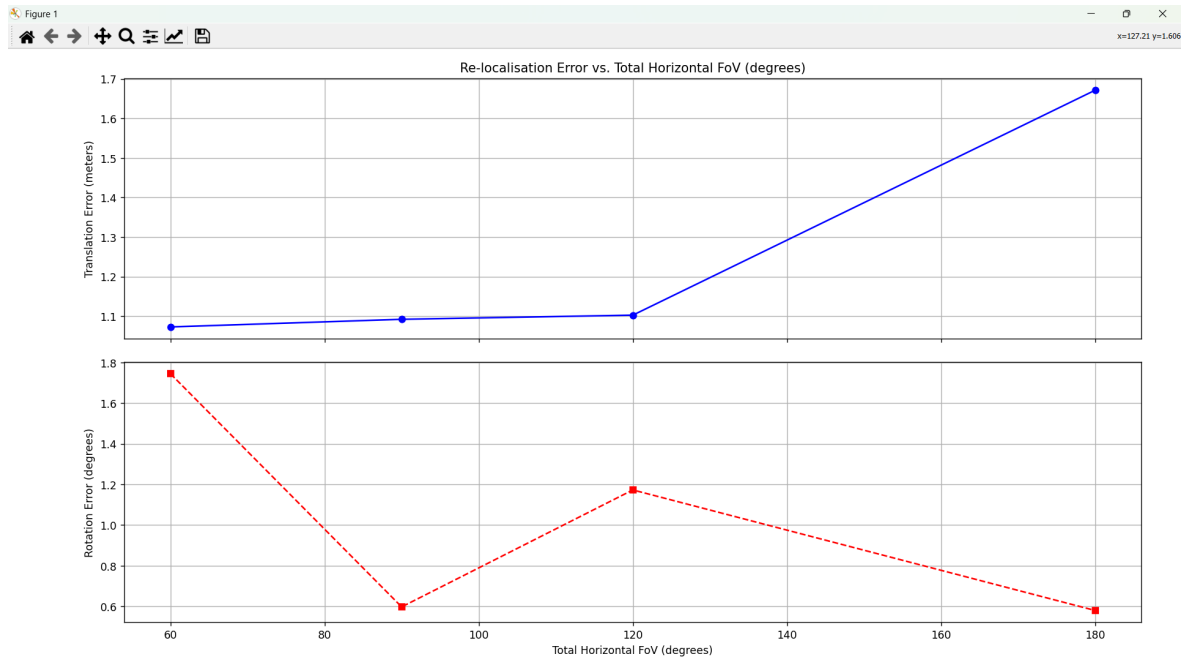


Figure 7.3: Re-localisation Translation and Rotation Error vs. Total Horizontal FoV.

7.4 Impact of Noise

Finally, Figure 7.4 displays the errors when varying the standard deviation of Gaussian noise added to the query point cloud coordinates, with other parameters held at default values (Range=40m, FoV=120°, Voxel=0.3m).

As seen in Figure 7.4, the translation error remains relatively stable for noise standard deviations up to 0.2m, increasing noticeably only at 0.3m. The rotation error appears remarkably robust across the tested noise levels in this experiment. This suggests that the point-to-plane ICP implementation used exhibits reasonable tolerance to moderate levels of coordinate noise, although performance eventually degrades as noise becomes excessive.

These results provide quantitative insights into how individual simulated sensor limitations affect the accuracy of ICP-based re-localisation, forming the basis for the discussion and conclusions presented later in this report.

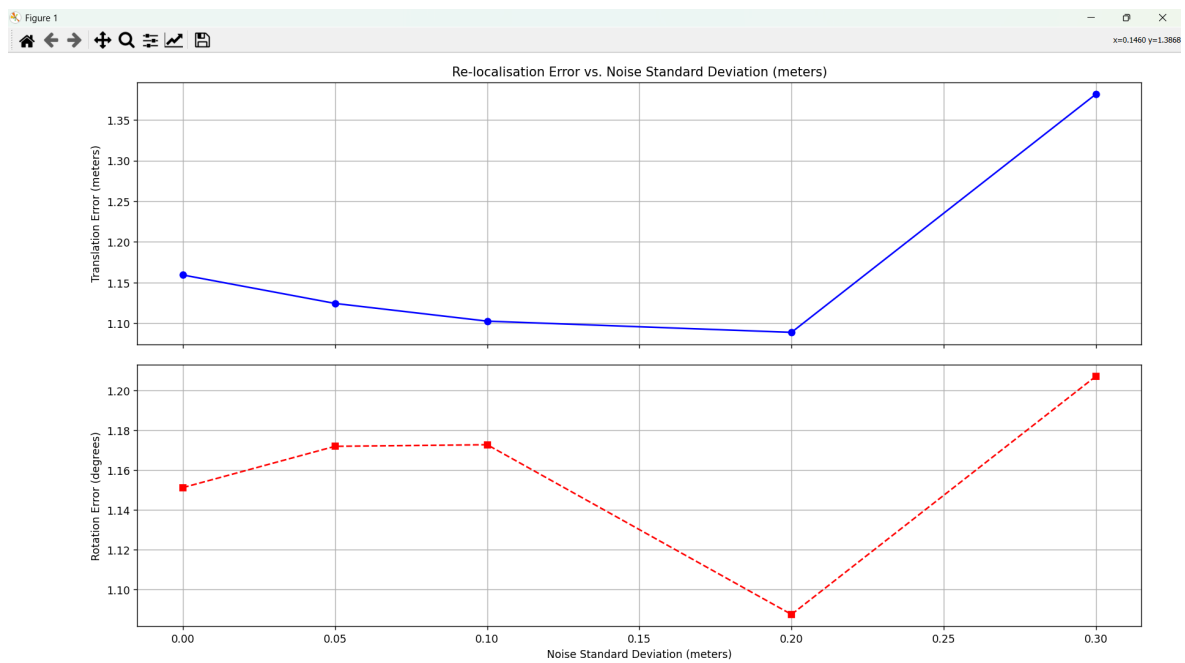


Figure 7.4: Re-localisation Translation and Rotation Error vs. Noise Standard Deviation.

Chapter 8: Summary and Conclusions

This project investigated the feasibility of employing LiDAR-based re-localisation techniques, commonly used in high-resource autonomous vehicle systems, on platforms characteristic of mobile and IoT devices with significant resource constraints. Recognizing the challenges in directly testing on diverse low-cost hardware, this study adopted a simulation-based methodology. High-quality point cloud data from the KITTI raw dataset [4] was systematically degraded to emulate the limitations of lower-cost sensors, specifically focusing on reduced sensing range, narrower field of view (FoV), decreased point density (via voxel downsampling), and increased measurement noise. The performance of a standard point-to-plane Iterative Closest Point (ICP) algorithm [9] was then evaluated for its ability to re-localise these degraded scans within a map constructed from the original, high-quality data. Accuracy was quantified by comparing the ICP-estimated pose against ground truth poses derived from the dataset's OXTS measurements, focusing on translation and rotation errors.

The experimental results, obtained through systematic variation of individual degradation parameters using a fixed random seed for reproducibility, and visualized in Figures 7.1 to 7.4, revealed several key insights. The impact of downsampling (voxel size) on translation error proved surprisingly non-monotonic (Figure 7.1), with a dramatic, unexpected drop in error at 0.5m voxel size, followed by increases and then decreases again at very large voxel sizes. This suggests potential ICP instability or sensitivity to specific geometric features remaining at different sparsity levels for this scenario. Rotation error, however, remained relatively stable across different voxel sizes. Reducing the maximum sensor range showed stable translation error between 30m and 60m, but again, an unexpected drop occurred at 15m before increasing slightly at 5m (Figure 7.2). Rotation error, conversely, remained low for ranges down to 15m but increased sharply at 5m, clearly indicating the challenge short-range sensors face in orientation estimation. Narrowing the field of view (Figure 7.3) generally decreased translation error down to 60°, suggesting focused data might be beneficial for ICP's positional estimate here. The rotation error showed a more complex trend, being lowest in the widest (180°) and narrower (90°) FoV, but increasing significantly at the narrowest (60°) setting. The system showed reasonable robustness to moderate Gaussian noise (Figure 7.4), with the translation error remaining relatively stable up to the 0.2m standard deviation before increasing at 0.3m. The rotation error was remarkably stable for all noise levels tested.

Based on these findings, several conclusions can be drawn regarding the feasibility of using standard ICP-based LiDAR re-localisation on resource-constrained devices. Although ICP can still converge with significantly degraded data, achieving consistently high accuracy is a challenge. The simulations highlight that orientation estimation is particularly vulnerable to very short sensor ranges (Figure 7.2) and very narrow field of view (Figure 7.3). The erratic performance observed under varying levels of downsampling (Figure 7.1) suggests that simply having fewer points is not always predictably worse for ICP translation, and the result likely depends heavily on the specific environment structure and the features preserved after downsampling. Moderate levels of noise appear manageable, but performance degrades at higher levels (Figure 7.4). This implies that careful co-design of algorithms and sensor characteristics (e.g., prioritizing sufficient range and FoV over raw density for certain applications) might be necessary. The study confirms that directly applying algorithms tuned for dense, clean data requires significant adaptation or acceptance of lower accuracy when faced with data characteristic of low-cost sensors.

This work contributes a quantitative baseline assessment using a reproducible simulation methodology on a standard dataset. However, certain limitations should be acknowledged. The experiments were carried out on a single KITTI sequence, representing an outdoor driving environment; performance in diverse indoor scenarios requires separate investigation. The simulation of sensor

degradation, while systematic, remains an approximation of real-world low-cost sensor behavior. Furthermore, only the ICP algorithm was evaluated; other registration techniques, such as NDT or feature-based methods, might exhibit different sensitivities to simulated degradations.

Future work could extend this simulation approach by testing combined degradation scenarios, evaluating performance over longer trajectories to analyze drift accumulation (odometry), exploring alternative registration algorithms (like NDT) and applying the methodology to different datasets representing varied environments (e.g., indoor scenes if suitable public datasets with ground truth become available).

Regarding ethical considerations, the project relied solely on publicly available data collected in public spaces, and the processing focused on anonymized geometric data, minimizing privacy concerns. Any future extension involving real-world deployment, particularly indoors, would require careful consideration of data privacy and user consent.

In conclusion, this project provides valuable quantitative insights into the performance trade-offs associated with deploying LiDAR re-localization on resource-constrained platforms. By simulating sensor limitations on high-quality benchmark data, it highlights the significant impact of reduced range, FoV, density, and increased noise (as shown in Figures 7.1 - 7.4) on the accuracy achievable with standard ICP, thus informing the design and expectations for future mobile and IoT localization systems.

Chapter 9: Future Work

While this project successfully established a simulation framework and provided quantitative insights into the feasibility of ICP-based LiDAR re-localisation under various sensor constraints, several avenues for future work remain open. These represent potential extensions, refinements, and alternative approaches that were beyond the scope or time constraints of the current investigation but could offer valuable further understanding.

9.1 Refining Simulation and Experimentation

- **Combined Degradation Scenarios:** The experiments conducted primarily varied one degradation parameter at a time. A crucial next step would be to simulate sensors with multiple limitations simultaneously (e.g., short range *and* high noise *and* low density *and* narrow FoV). This would provide a more realistic assessment of performance under challenging, low-cost sensor profiles.
- **Extended Odometry/Sequence Analysis:** The frame-to-frame analysis was limited to a short sequence (10 pairs). Extending this odometry simulation over the entire drive sequence (or significant portions) would allow for a more thorough analysis of error accumulation and long-term drift under different degradation levels.
- **Different KITTI Sequences/Environments:** The current results are based on a single residential sequence. Repeating the re-localisation and odometry experiments on other KITTI sequences (e.g., highway, city center) would test the generality of the findings across different environment types and complexities.
- **Indoor Datasets:** A significant limitation was the use of an outdoor dataset. Future work should ideally incorporate publicly available indoor LiDAR datasets (if suitable ones with ground truth exist) or involve collecting new data in representative indoor environments to specifically address the challenges unique to those spaces (e.g., featureless corridors, different clutter).
- **Improved Sensor Simulation Fidelity:** The current degradation models (range/FoV cut-offs, voxel downsampling, Gaussian noise) are approximations. More sophisticated simulation could involve modeling specific scan patterns of low-cost sensors (e.g., non-repetitive scans), more realistic noise models (e.g., range-dependent noise), or learned models of specific sensor behaviours [30].
- **Varying Map Density:** The current map was built from original data and moderately downsampled. Investigating how re-localisation performs when matching against maps built with different densities (sparser or denser) could yield further insights.

9.2 Exploring Alternative Algorithms

This project focused on evaluating the standard point-to-plane ICP algorithm. Given its observed sensitivities, particularly under heavy degradation, exploring alternative algorithms is a key future direction:

- **Normal Distributions Transform (NDT):** Evaluating NDT [10] under the same degradation scenarios would be valuable, as its statistical representation might offer better robustness to noise and density variations, although potentially at the cost of fine-grained accuracy or different computational trade-offs [13].
- **Feature-Based Methods:** Investigating the robustness of feature extraction (e.g., from LeGO-LOAM [5]) on the degraded data and evaluating the performance of feature-based matching could provide an alternative, potentially more efficient approach if features remain reliable.
- **Robust ICP Variants:** Testing other ICP variants specifically designed for outlier rejection or handling partial overlap (e.g., Trimmed ICP, G-ICP [6]) might show improved performance compared to the standard point-to-plane version used here.
- **Deep Learning Approaches:** While computationally more demanding, exploring lightweight deep learning models for feature extraction, scan matching, or place recognition (inspired by [7, 16] but optimized for mobile platforms) could offer significant robustness improvements, especially if hardware acceleration is available.

9.3 Sensor Fusion and System Integration

- **LiDAR-IMU Fusion:** Integrating Inertial Measurement Unit (IMU) data is crucial for practical mobile robotics. IMU data can provide high-frequency motion priors, aid in motion distortion correction (deskewing), and help constrain the localisation solution, especially during fast rotations or in feature-poor areas where LiDAR-only methods struggle. Evaluating tightly-coupled LiDAR-Inertial Odometry (LIO) algorithms (like simplified versions of [31]) under degradation would be a valuable next step.
- **Visual-LiDAR Fusion:** Combining LiDAR with visual data from cameras could leverage the strengths of both sensors – geometric accuracy from LiDAR and texture information from cameras – potentially leading to more robust localisation than either sensor alone, although adding complexity.
- **Hardware Profiling/Deployment:** While this project focused on simulation, a logical follow-up would involve implementing the most promising degraded simulation setup (e.g., ICP or NDT with specific parameters) on actual resource-constrained hardware (like a Raspberry Pi or Nvidia Jetson) and profiling its real-world performance (CPU/memory usage, latency, energy consumption).

This project provides a foundation by quantifying the performance of a baseline algorithm under simulated constraints. The avenues outlined above offer numerous possibilities for building upon this work, aiming towards robust, efficient, and accurate LiDAR-based localisation suitable for the unique challenges of mobile and IoT devices.

9.4 Conclusion

This project set out to assess the feasibility of using LiDAR-based re-localisation, a technique common in autonomous vehicles, on resource-constrained mobile and IoT devices. The core aim was achieved by developing and executing a simulation pipeline that systematically degraded high-quality KITTI dataset point clouds to mimic the limitations (range, FoV, density, noise) of lower-cost sensors and evaluating the performance of a standard ICP algorithm under these conditions.

The main results demonstrated quantitatively how these simulated degradations impact re-localisation accuracy. It was found that while ICP can still function with significantly degraded data, accuracy suffers noticeably, particularly with very short sensor ranges and narrow fields of view (which heavily impacted rotation estimation) or high levels of noise (which significantly impacted translation estimation). The effect of reduced point density (downsampling) was found to be less predictable for ICP in this specific scenario.

The primary achievement of this work is the successful implementation of this simulation framework and the generation of reproducible, quantitative results that characterize the performance baseline of ICP under these specific constraints. This provides valuable, data-driven insights into the challenges and trade-offs involved in deploying LiDAR re-localisation on devices with limited sensor capabilities, fulfilling the project's core objective of conducting a simulation-based feasibility assessment. While further work is needed, particularly regarding combined degradations, alternative algorithms, and real-world hardware testing, this project establishes a solid foundation and methodology for such future investigations.

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